

Database Architecture — Best Invest Properties

Date: 25 September 2026

Storage: Bubble database

Participating systems: Bubble, Make, OpenAI API

1. Purpose and principles

The model must support the three real MVP loops:

- an investor finds, compares, saves and enquires about a specific available unit;
- a developer gets verified, submits a project and maintains price/availability;
- an admin checks documents, publishes content and controls introductions, the score and the AI narrative.

Principles:

1. **Bubble is the source of truth.** The Make Data Store is not a business database.
2. **Unit is the public listing object.** Project describes the building/complex, Unit Type the repeated configuration, Unit the specific offer.
3. **A single property does not get its own table.** It is a project with one type and one unit.
4. **Finance is deterministic.** AI never writes price, rent, yield, score or tax.
5. **Published data is versioned.** Critical changes have a history and an author.
6. **Privacy by default.** New types are created private; only whitelisted fields of published listings are public.
7. **Minimal denormalisation for Bubble.** Fields needed by privacy rules and frequent searches are duplicated on the protected record.
8. **Idempotent integrations.** Every asynchronous process has an Integration Job and a unique idempotency key.

Bubble privacy rules run on the server and must be the primary mechanism that keeps data out of the browser. Visually hiding a group or button is not access control.

2. Conceptual schema

erDiagram

```
USER ||--o| INVESTOR_PROFILE : has
USER ||--o{ COMPANY_MEMBERSHIP : belongs_via
DEVELOPER_COMPANY ||--o{ COMPANY_MEMBERSHIP : has
USER ||--o{ DEVELOPER_APPLICATION : submits
DEVELOPER_COMPANY ||--o{ DEVELOPER_APPLICATION : concerns
DEVELOPER_APPLICATION ||--o{ VERIFICATION_DOCUMENT : includes
```

```
COUNTRY_CONFIG ||--o{ PROJECT : contains
DEVELOPER_COMPANY ||--o{ PROJECT : owns
PROJECT ||--o{ UNIT_TYPE : defines
UNIT_TYPE ||--o{ UNIT : instantiates
PROJECT ||--o{ MEDIA_ASSET : has
UNIT_TYPE ||--o{ MEDIA_ASSET : has
```

```
SCORE_MODEL_VERSION ||--o{ SCORE_WEIGHT : contains
SCORE_CRITERION ||--o{ SCORE_WEIGHT : weighted_by
UNIT ||--o{ LISTING_SCORE : receives
SCORE_MODEL_VERSION ||--o{ LISTING_SCORE : produced_by
LISTING_SCORE ||--o{ SCORE_COMPONENT : contains
UNIT ||--o{ AI_ANALYSIS : explained_by
```

```
USER ||--o{ SAVED_UNIT : saves
USER ||--o{ SAVED_SEARCH : owns
USER ||--o{ CALCULATOR_SCENARIO : owns
USER ||--o{ ENQUIRY : creates
UNIT ||--o{ ENQUIRY : concerns
ENQUIRY ||--o{ ENQUIRY_STATUS_EVENT : has
ENQUIRY ||--o| CONTACT_RELEASE : may_create
```

```
PROJECT ||--o{ CHANGE_REQUEST : receives
CHANGE_REQUEST ||--o{ CHANGE_ITEM : contains
USER ||--o{ CONSENT_RECORD : grants
USER ||--o{ NOTIFICATION : receives
INTEGRATION_JOB ||--o{ AUDIT_EVENT : emits
```

3. Naming rules in Bubble

- Data types: singular English, Pascal Case: `Developer Company`, `Score Model Version`.
- Fields: snake_case English: `publication_status`, `price_eur`.
- Option sets: Pascal Case: `Project Status`; options are stable lower-case machine keys.
- Every business type has: `public_id` (text for URLs/references), `is_archived`, `created_by_user`, and `version_number` where needed.
- The Bubble `unique id` is used internally only; `public_id` is what goes outside.

- Money in the MVP: numeric fields with the `_eur` suffix; `currency_code` is stored even if the value is always EUR.
- Percentages: store the decimal fraction (`0.0725`), display as `7.25%`.
- Time: all timestamps in UTC; timezone is used only for display.

4. Option Sets

Option Sets are only for non-secret, rarely changing values. Bubble states explicitly that option sets are part of the app code, are not protected by privacy rules and are unsuitable for sensitive data.

Option Set	MVP values
User Role	<code>investor</code> , <code>developer</code> , <code>admin</code> (roles <code>senior_admin</code> , <code>support</code> reserved for later)
Account Status	<code>pending_email</code> , <code>active</code> , <code>suspended</code> , <code>closed</code>
Application Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>more_info_required</code> , <code>approved</code> , <code>rejected</code> , <code>withdrawn</code>
Project Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>changes_requested</code> , <code>approved</code> , <code>published</code> , <code>paused</code> , <code>sold_out</code> , <code>archived</code> , <code>rejected</code>
Publication Status	<code>not_published</code> , <code>published</code> , <code>hidden</code> , <code>archived</code>
Unit Availability	<code>available</code> , <code>reserved</code> , <code>sold</code> , <code>withdrawn</code>
Financial Input Source	<code>developer_claim</code> , <code>portal_comparable</code> , <code>analyst_verified</code> , <code>platform_default</code> , <code>calculated</code>
Analysis Fact Tag	<code>source</code> , <code>developer</code> , <code>estimate</code> , <code>gap</code>
Change Request Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>queried</code> , <code>approved</code> , <code>rejected</code> , <code>applied</code> , <code>cancelled</code>
Enquiry Status	<code>submitted</code> , <code>screening</code> , <code>on_hold</code> , <code>declined</code> , <code>approved_for_intro</code> , <code>introduced</code> , <code>developer_responded</code> , <code>qualified</code> , <code>closed_won</code> , <code>closed_lost</code> , <code>cancelled</code>
Score Verdict	<code>strong</code> , <code>good</code> , <code>watch</code> , <code>high_risk</code> , <code>not_scored</code>
Review Status	<code>not_required</code> , <code>pending</code> , <code>approved</code> , <code>changes_requested</code> , <code>rejected</code>
Analysis Status	<code>queued</code> , <code>generating</code> , <code>generated</code> , <code>failed</code> , <code>stale</code> , <code>published</code>
Job Status	<code>queued</code> , <code>processing</code> , <code>succeeded</code> , <code>retry_wait</code> , <code>failed</code> , <code>dead_letter</code> , <code>cancelled</code>
Notification Status	<code>queued</code> , <code>sending</code> , <code>sent</code> , <code>failed</code> , <code>cancelled</code>
Notification Channel	<code>in_app</code> , <code>email</code>
Document Kind	<code>company_registration</code> , <code>licence</code> , <code>ownership</code> , <code>planning</code> , <code>title</code> , <code>brochure</code> , <code>floor_plan</code> , <code>legal</code> , <code>other</code>

Option Set	MVP values
Financing Mode	cash , mortgage
Strategy	long_term_rental , short_term_rental , capital_growth , mixed

Country, city, amenities, document requirements, scoring criteria, tax rules and legal documents are **not** option sets: they must be editable without a deploy and have a history.

Unit Availability . A developer may set only `available` , `reserved` or `sold` , and every such change goes through a Change Request with approval. Only an admin sets `withdrawn` . This must be enforced in the backend workflow whitelist, not in the UI.

Enquiry Status . The set of eleven values is **canonical**. Role-specific labels are a separate lookup, not additional statuses.

Role-specific status labels

Canonical status	Admin	Investor	Developer
<code>submitted</code>	New	Request received	not shown
<code>screening</code>	Qualifying	Awaiting analysis	not shown
<code>qualified</code>	Hot lead / Awaiting you	Awaiting admin approval	Awaiting admin approval
<code>on_hold</code>	On hold	Under review	On hold
<code>declined</code>	Declined	Not available	Filtered out by Best Invest
<code>approved_for_intro</code>	Connecting	Introduction being prepared	Introduction being prepared
<code>introduced</code>	Connected	Developer contacted	Introduced
<code>developer_responded</code>	In progress	Developer responded	Contacted
<code>closed_won</code>	Converted	Completed	Converted
<code>closed_lost</code>	Closed	Closed	Closed
<code>cancelled</code>	Cancelled	Cancelled	Cancelled

Implementation: the `Enquiry Status` Option Set with three display attributes (`label_admin` , `label_investor` , `label_developer`). An empty label means the record is not shown to that role at all — this is enforced by a privacy rule, not just hidden in the UI.

AWAITING YOU is not a status. It is the admin label meaning a lead in `qualified` is waiting for their decision. Likewise, "Hot lead" is a label for the same status, not a separate stage.

5. Data types

5.1 Identity and access

User (Bubble built-in)

Field	Type	Required	Note
<code>public_id</code>	text	yes	<code>usr_...</code> , not the email
<code>full_name</code>	text	yes	do not use as an identifier

Field	Type	Required	Note
phone_e164	text	no	private
country_of_residence	Country Config	no	private
roles	list of User Role	yes	at least one role after activation
account_status	Account Status	yes	route guard
email_verified	yes/no	yes	false by default
locale	text	yes	en in the MVP
last_login_at	date	no	audit
terms_version_accepted	text	no	UI cache; source is the Consent Record
privacy_version_accepted	text	no	UI cache
admin_permission_keys	list of text	no	admin only; do not rely on the role alone

Do not store the password in a custom field. Use Bubble authentication.

Investor Profile

`user`, `budget_min_eur`, `budget_max_eur`, `target_gross_yield`, `target_net_yield`, `time_horizon_months`, `countries` (list Country Config), `strategies` (list Strategy), `bedrooms_min`, `profile_completed_at`, `marketing_email_opt_in`, `match_email_frequency`, `assigned_admin_user`.

One active Investor Profile per User. Uniqueness is checked by a backend workflow before creation.

Developer Company

`public_id`, `legal_name`, `trading_name`, `registration_number`, `registration_country`, `registered_address`, `website_url`, `vat_number`, `verification_status`, `verified_at`, `verified_by_user`, `relationship_owner_user`, `anonymity_default`, `is_suspended`, `is_archived`.

The Company owns the Project. Never link a Project directly to a single developer User.

Company Membership

`company`, `user`, `member_role_key`, `can_manage_projects`, `can_change_prices`, `can_view_leads`, `status`, `invited_at`, `joined_at`.

The MVP creates one membership, but the structure is ready for a team.

Developer Application

`public_id`, `company`, `submitted_by_user`, `status`, `markets` (list Country Config), `years_active`, `completed_projects_count`, `submitted_at`, `review_started_at`, `decided_at`, `decision_by_user`, `decision_reason_private`, `decision_message_public`, `current_review_round`, `terms_version_accepted`.

Verification Document

application, company, country, document_kind, file, original_filename, mime_type, size_bytes, expires_at, verification_status, review_note_private, review_note_public, reviewed_by_user, reviewed_at, is_current.

The file is always private and attached to the Verification Document. Removing the URL does not delete the file; the delete workflow must first run the Bubble action "Delete an uploaded file".

5.2 Market configuration

Country Config

iso2, name, currency_code, is_live, coming_soon, default_vacancy_rate, default_management_fee_rate, default_maintenance_rate, default_insurance_annual_eur, legal_disclaimer, calculator_config_version, effective_from, effective_to.

Do not store all tax rules as one text. Use Cost Rule for a transparent calculation.

Comparable-listing sample thresholds. Stored here rather than globally, because market liquidity differs: Nicosia has fewer listings than Málaga, and a single threshold would be either too strict or meaningless.

Field	Type	Default	Purpose
min_comparable_listings	number	5	below this the sample cannot be approved at all
sufficient_comparable_listings	number	10	below this the sample is approved but flagged "thin sample"
max_comparable_age_days	number	90	after this the sample is stale and needs refreshing

Reasoning for the defaults: a median of four listings is not a market figure — one outlier moves it too much, so 5 is the absolute minimum. From 10 listings the median is stable, so the 5–9 band stays usable but explicitly flagged. 90 days is a quarter: long enough not to re-collect every week, short enough for the estimate to keep up with the market.

The values are **configurable per country** by the admin without code changes.

Cost Rule

country, rule_key, label, applies_to_property_type, applies_to_new_build, calculation_method (flat, percent_price, banded, manual), rate, fixed_amount_eur, bands_json, effective_from, effective_to, source_url, source_checked_at, approved_by_user, version_number.

bands_json is for configuration only; the final calculated amounts are stored in separate numeric fields on the Financial Snapshot.

Document Requirement

country, subject_type (developer, project), document_kind, required_for_account_creation, required_for_submission, required_for_publication, expires, expiry_months, instructions, is_active, effective_from.

The `required_for_account_creation` field separates documents needed to open a developer account (company registration, licence) from those needed only before a project is published (building permit, escrow). This expresses the three levels on the application screen: `REQUIRED` / `TO CONFIRM` / `OPTIONAL`.

The matrix is **fully configurable** by the admin per country and is not hard-coded.

Amenity

key, label, scope (project, unit_type, unit), countries, is_filterable, sort_order, is_active.

5.3 Property catalogue

Project

public_id, company, country, city, area_name, address_private, map_lat_public_rounded, map_lng_public_rounded, name_internal, name_public, project_type, description_source, description_public, completion_date, declared_unit_count, status, publication_status, published_at, published_by_user, cover_media, amenities, developer_visible_publicly, current_content_version, is_archived.

Do not expose `address_private` publicly if the business wants to anonymise the exact location. Country/city/area are enough for search cards.

Developer anonymity. The developer's name and contacts are not shown in the catalogue or on the property page; the public label is "Introduced by Best Invest". So in the MVP `developer_visible_publicly` is **always no**: the field stays in the schema for a future policy change but is not shown in the UI and cannot be edited by the developer. The privacy rule on Project must not return `company` to an anonymous request at all — do not rely on the UI not showing it.

Unit Type

public_id, project, type_code, display_name, bedrooms, bathrooms, indoor_area_m2, outdoor_area_m2, floor_plan_media, default_expected_monthly_rent_eur, features, is_active.

Unit

This is the central listing record.

Field	Type	Purpose
public_id	text	URL and external references
project	Project	parent project
unit_type	Unit Type	standard configuration

Field	Type	Purpose
unit_number_internal	text	private until introduction if needed
display_reference	text	public number/name
floor	number	filter/details
orientation	text	optional
price_eur	number	current approved price
expected_monthly_rent_eur	number	cache of the approved Financial Input used for scoring
rent_source_type	Financial Input Source	source cache for quick display
rent_source_reference	text	traceability
has_pending_change	yes/no	a Change Request is in progress — the portal row shows "pending approval"
availability	Unit Availability	current state
reservation_expires_at	date	if reserved
gross_yield	number	cache of the deterministic formula
net_yield	number	cache of the deterministic formula
investment_score	number	current published score
score_verdict	Score Verdict	badge
current_score_record	Listing Score	version/components
completion_date_cached	date	for fast search
country_cached	Country Config	for privacy/search without a deep chain
city_cached	text	search
bedrooms_cached	number	search
area_m2_cached	number	search
project_type_cached	text	search
strategy_keys	list of Strategy	search
publication_status	text	only published is public
published_at	date	audit
last_financial_recalc_at	date	freshness

Cached fields are updated by the atomic backend workflow `rebuild_unit_public_snapshot`. Do not build the public search over many nested relations.

Media Asset

`public_id`, `project`, `unit_type`, `unit`, `kind`, `image`, `file`, `is_private`, `is_cover`, `sort_order`, `caption`, `mime_type`, `size_bytes`, `width_px`, `height_px`, `upload_status`, `uploaded_by_user`, `review_status`, `rights_confirmed`, `is_archived`.

Public photos/brochures and private legal documents are different records with different privacy rules.

Project Content Version

```
project, version_number, snapshot_json, created_by_user, created_at, reason,
source_change_request, is_published_version.
```

Stores a reconstruction of the published state. For frequently needed fields the live value stays on Project/Unit.

5.4 Finance and score

Financial Snapshot

```
unit, version_number, price_eur, monthly_rent_eur, annual_gross_rent_eur, occupancy_rate,
annual_effective_rent_eur, annual_management_eur, annual_maintenance_eur,
annual_insurance_eur, annual_property_tax_eur, annual_other_costs_eur, purchase_costs_eur,
annual_net_income_eur, gross_yield, net_yield, country_config_version, calculated_at,
calculated_by_workflow_version, is_current.
```

Base MVP formulas:

```
annual_gross_rent = monthly_rent × 12
annual_effective_rent = annual_gross_rent × occupancy_rate
gross_yield = annual_gross_rent / price
annual_net_income = annual_effective_rent
    - management - maintenance - insurance - property_tax - other_costs
net_yield = annual_net_income / (price + purchase_costs)
```

A formula change creates a new version rather than rewriting history. The inputs to the formulas are approved Financial Inputs (rent comes from comparable listings via Make) and the Cost Rules for each country.

Financial Input

The Project Review screen shows a panel where every figure has its own source, and three different rent values side by side (claimed by the developer, Best Invest's comparable estimate, and the one actually used for scoring). So each input value is stored as a separate record.

Field	Type	Required	Purpose
unit	Unit	yes	which unit it belongs to
input_key	text	yes	expected_annual_rent, recurring_costs, vacancy_allowance, ...
label	text	yes	label in the review
value_number	number	yes	value
unit_of_measure	text	no	eur_per_month, eur_per_year, rate

Field	Type	Required	Purpose
source	Financial Input Source	yes	developer_claim / portal_comparable / analyst_verified / platform_default / calculated
source_reference	text	no	"Larnaca district, 14 lettings"
comparable_set	Rental Comparable Set	no	required for portal_comparable — all sample details live there
verified_at	date	no	date checked by the admin
verified_by_user	User	no	who checked
is_used_for_scoring	yes/no	yes	only one active value per input_key
review_statuses	Review Status	yes	pending until approved by the admin
approved_by_user	User	no	who approved
approved_at	date	no	when
superseded_by	Financial Input	no	history instead of overwrite

Rules:

- only values with review_status = approved reach the public listing;
- the Financial Snapshot is built **only** from approved Financial Inputs;
- a value with source = developer_claim can never have is_used_for_scoring = yes without separate analyst approval (BR-02);
- source = portal_comparable requires a comparable_set with review_status = approved — without an approved sample the value cannot be approved;
- for rent there are always at least two records in the database: the developer's claim (developer_claim) and the value derived from comparable listings (portal_comparable). These are what the Project Review screen shows side by side;
- developer_claim never turns into portal_comparable automatically — they are different records with different provenance;
- a record is not edited: a new value creates a new record and sets superseded_by on the old one.

Rental Comparable Set

The result of collecting comparable rental listings via Make. One record = one sample for one parameter set on a given date.

Field	Type	Required	Note
public_id	text	yes	
provider	Data Provider	yes	portal or market-data provider
retrieved_at	date	yes	date the data was retrieved

Field	Type	Required	Note
country	Country Config	yes	normalisation parameter
city	text	yes	normalisation parameter
area_name	text	no	district
property_type	text	yes	normalisation parameter
bedrooms	number	yes	normalisation parameter
area_m2_min	number	no	sample floor-area bound
area_m2_max	number	no	sample floor-area bound
listings_count	number	yes	number of comparable listings
rent_min_eur	number	yes	lower end of the range
rent_median_eur	number	yes	median — basis of the base estimate
rent_max_eur	number	yes	upper end of the range
collection_method	text	yes	api / feed / manual_import / manual_entry
integration_job	Integration Job	no	empty for manual entry
review_status	Review Status	yes	pending until checked by the admin
reviewed_by_user	User	no	
reviewed_at	date	no	
is_current	yes/no	yes	current sample for this parameter set

Rules:

- a sample with `review_status ≠ approved` cannot be a source for a Financial Input and is not shown to the investor;
- `listings_count` below `Country Config.min_comparable_listings` (default **5**) **blocks approval** — an estimate from four listings is not a market figure;
- `listings_count` between the minimum and `sufficient_comparable_listings` (default **10**) is approved but flagged **thin sample**: the flag is shown to the admin on the review screen and added for the investor as an Analysis Fact tagged `estimate`;
- a sample older than `max_comparable_age_days` (default **90**) gets `is_current = no` and cannot be the source of a new Financial Input; existing published values do not disappear but are queued for refresh;
- a new sample does not overwrite the previous one: the old one gets `is_current = no`, so historical scores remain reproducible;
- a manual fallback (`collection_method = manual_import|manual_entry`) is stored with **the same type** and the same set of fields, so the provenance of the figure stays visible on the review screen.

Data Provider

Reference table of approved sources. Not an option set: the list changes without a deploy and needs a history.

`key`, `name`, `country`, `access_method` (`api` / `feed` / `manual`), `terms_reference`, `is_active`, `approved_by_user`, `approved_at`, `last_successful_fetch_at`, `notes`.

The legal basis for using each portal's data (API terms, data licence) must be checked before connecting it. The `terms_reference` field stores a link to the relevant document.

Analysis Fact

The "Sources, assumptions & gaps" block on the analysis and review screens.

`unit`, `tag` (Analysis Fact Tag), `body`, `source_url`, `source_checked_at`, `sort_order`, `is_active`, `created_by_user`.

The `gap` tag marks a known absence of data and must be visible to the investor — it is a legally significant part of the disclosure.

Score Criterion

`key`, `label`, `description`, `direction` (`higher_better`, `lower_better`, `target_range`), `input_source`, `calculation_rule_version`, `is_active`, `sort_order`.

Seed — five criteria. This set is used on the Analysis, Compare, Score Editor and Project Review screens:

sort_order	key	label	max_points	weight_fraction	direction
1	income	Rental Income & Net Yield	30	0.30	higher_better
2	demand	Rental Demand & Tenant Quality	20	0.20	higher_better
3	value	Purchase Value & Market Position	20	0.20	higher_better
4	growth	Growth & Resale Potential	15	0.15	higher_better
5	risk	Risk & Investor Protection	15	0.15	higher_better

For `risk`, more points means **lower** risk, so `direction` is `higher_better`, not `lower_better`. The scale label is mandatory in the UI.

Normalisation rubric. Each criterion gets a `rating` of 0-10, then `weighted_points` = `rating` / 10 × `weight`. The full scales are in FR-03b of the project specification. For the `income` criterion the rating is derived automatically from net yield bands; for the other four it is built from sub-criteria assessed by the analyst.

Score Sub-criterion

Reference table of sub-criteria for the four manually assessed criteria.

`criterion`, `key`, `label`, `max_points`, `sort_order`, `guidance`, `is_active`.

Seed:

criterion	key	label	max_points
demand	market_activity	Market activity and number of comparable listings	3
demand	year_round	Year-round demand	3
demand	tenant_mix	Diversity of potential tenants	2

criteria	key	label	max_points
demand	seasonality	Seasonality and vacancy risk	2
growth	price_trend	Price trend	4
growth	liquidity	Liquidity and number of transactions	3
growth	infrastructure	Infrastructure and economic factors	2
growth	data_quality	Recency and completeness of data	1
risk	legal_title	Legal status / title	3
risk	permits	Building permits and documents	2
risk	payment_protection	Payment / escrow protection	2
risk	developer_check	Developer check	2
risk	stage_risk	Construction stage risk	1

The `value` criterion has no sub-criteria: its rating is derived from how far the price per m² deviates from the median (see Market Benchmark).

The `max_points` of each criterion's sub-criteria must add up to **10**. A backend workflow checks this when the reference table is saved.

Score Sub-component

The actual assessment of a sub-criterion for a specific Listing Score.

`listing_score`, `sub_criterion`, `points_awarded`, `note`, `assessed_by_user`, `assessed_at`.

`points_awarded` may not exceed `sub_criterion.max_points`. The sum of `points_awarded` within a criterion gives its `rating`.

Market Benchmark

Needed for the `Purchase Value & Market Position` criterion, which compares the price per m² with the district median.

Field	Type	Required	Note
country	Country Config	yes	
city	text	yes	
area_name	text	no	empty = whole city
property_type	text	no	
benchmark_key	text	yes	<code>median_price_per_m2_new_build</code>
value_number	number	yes	€/m ²
sample_size	number	no	number of properties in the sample
source_type	text	yes	public listings / transaction register
source_reference	text	no	
checked_at	date	yes	date checked
checked_by_user	User	yes	

Field	Type	Required	Note
effective_from	date	yes	
effective_to	date	no	

Benchmarks are versioned by date, not overwritten: historical scores must stay reproducible.

Score Model Version

`public_id`, `version_number`, `country` (empty = global), `strategy` (optional), `status` (draft, active, retired), `effective_from`, `created_by_user`, `approved_by_user`, `approved_at`, `change_reason`.

Score Weight

`model_version`, `criterion`, `weight_fraction`, `max_points`. For the active model the sum of `weight_fraction` must equal 1.0; Save is disabled otherwise.

The weights editor works in percentages (0-100 per criterion, total = 100). The database stores the decimal fraction (0.30); conversion happens in the UI. There is **no** hard cap on a single criterion's weight: the only rule the backend workflow checks is that the active model's `weight_fraction` values add up to 1.0.

The UI shows a soft warning if any criterion exceeds **50%** — in that case one criterion effectively determines the whole score and the other four become decorative. The warning **does not block** saving.

The 50% threshold is a UI constant, not a database field: changing it needs no migration.

Listing Score

`unit`, `model_version`, `financial_snapshot`, `total_score`, `verdict`, `calculated_at`, `calculation_run_id`, `is_current`, `superseded_at`, `review_status`.

Score Component

`listing_score`, `criterion`, `raw_value_number`, `rating`, `weighted_points`, `source_type`, `source_record_id`, `explanation_deterministic`.

`rating` is a 0-10 score. `weighted_points` is calculated as `rating / 10 × criterion.weight_fraction × 100`.

Components must add up to the total score within a rounding tolerance of 0.01.

AI Analysis

`unit`, `listing_score`, `financial_snapshot`, `status`, `review_status`, `prompt_version`, `schema_version`, `model_id`, `input_snapshot_json`, `output_json`, `summary`, `strengths_json`, `risks_json`, `source_keys_json`, `openai_response_id`, `input_tokens`, `output_tokens`, `estimated_cost_usd`, `generated_at`, `reviewed_by_user`, `reviewed_at`, `published_at`, `failure_code`, `failure_message_private`.

Only `review_status = approved` is published, and only if the linked score/financial snapshot is still current.

5.5 Investor journey

Saved Unit

`user`, `unit`, `saved_at`, `is_active`, `removed_at`, `price_at_save_eur`, `availability_at_save`. Check for an active duplicate before creating.

Saved Search

`public_id`, `user`, `name`, `country`, `price_min_eur`, `price_max_eur`, `property_types`, `bedrooms_min`, `gross_yield_min`, `net_yield_min`, `strategies`, `completion_from`, `completion_to`, `alert_frequency`, `is_active`, `last_matched_at`.

Calculator Scenario

`public_id`, `user`, `unit`, `name`, `financing_mode`, `purchase_price_eur`, `deposit_eur`, `loan_amount_eur`, `interest_rate`, `term_years`, `monthly_rent_eur`, `occupancy_rate`, `management_fee_rate`, `purchase_costs_eur`, `annual_net_income_eur`, `cash_on_cash_return`, `formula_version`, `is_saved`, `created_at`.

Anonymous calculations are not stored. Saving requires login.

Enquiry

Field	Type	Note
<code>public_id</code>	text	ENQ-YYYY-...
<code>investor_user</code>	User	private
<code>investor_profile</code>	Investor Profile	snapshot source
<code>unit</code>	Unit	the specific subject of the enquiry
<code>project</code>	Project	denormalised
<code>developer_company</code>	Developer Company	denormalised
<code>status</code>	Enquiry Status	canonical status
<code>budget_snapshot_eur</code>	number	value at submit time
<code>criteria_snapshot_json</code>	text	immutable context
<code>consent_to_share_contact</code>	yes/no	separate consent
<code>fit_score</code>	number	deterministic/analyst; not AI without explanation
<code>fit_summary_private</code>	text	admin-only
<code>assigned_admin_user</code>	User	owner
<code>submitted_at</code>	date	SLA start
<code>admin_decision_at</code>	date	SLA
<code>developer_response_due_at</code>	date	48h
<code>closed_at</code>	date	lifecycle
<code>outcome_reason</code>	text	controlled values + note

Enquiry Status Event

`enquiry`, `from_status`, `to_status`, `actor_user`, `actor_role`, `occurred_at`, `reason_code`, `note_private`, `note_investor`, `note_developer`, `automation_job`.

A status never changes without a Status Event. Role-specific UI labels may differ, but there is one canonical status.

Contact Release

`enquiry`, `approved_by_user`, `approved_at`, `consent_record`, `investor_fields_released`, `developer_fields_released`, `delivery_job`, `delivered_at`, `revoked_at`, `revocation_reason`.

The developer privacy rule on User is never opened. The developer sees the permitted contact fields through the Contact Release / a separate projection, not through broad access to User.

5.6 Changes, consent, notification, audit

Change Request

`public_id`, `project`, `requested_by_user`, `company`, `status`, `request_type`, `submitted_at`, `assigned_admin_user`, `decision_at`, `decision_by_user`, `decision_reason_private`, `decision_message_public`, `applied_at`, `idempotency_key`.

Change Item

`change_request`, `target_type`, `target_public_id`, `field_key`, `old_value_text`, `new_value_text`, `old_value_number`, `new_value_number`, `old_value_date`, `new_value_date`, `old_value_option_key`, `new_value_option_key`, `review_status`, `review_note`, `applied_at`.

MVP whitelist of editable fields for a published project: `Unit.price_eur`, `Unit.availability`. Any other change triggers a full content review.

Both fields go through **prior** approval — there is no immediate availability change. Availability values a developer may set: `available`, `reserved`, `sold`. Only the admin sets `withdrawn`.

Locked fields, shown in the "LOCKED — ADMIN REVIEW REQUIRED" panel on the Project & Units screen: project name, location, completion date, unit mix, specification, media. A request to change them creates a Change Request with `request_type = content_review`.

Admin decisions on a Change Request in the queue: `approved`, `queried`, `rejected`. `queried` and `rejected` require a reason, which is stored in `decision_reason_private` and sent to the developer in `decision_message_public`.

Consent Record

`user`, `consent_type`, `granted`, `document_version`, `occurred_at`, `source_page`, `ip_hash`, `user_agent_hash`, `withdrawn_at`, `supersedes_record`.

Do not overwrite consent with a single yes/no without history.

Notification

`user`, `channel`, `template_key`, `entity_type`, `entity_public_id`, `title`, `body_preview`, `status`, `scheduled_at`, `sent_at`, `read_at`, `integration_job`, `failure_code`.

Legal Document Version

`document_type`, `country`, `language`, `version`, `status`, `effective_from`, `content`, `file`, `approved_by_user`, `approved_at`, `supersedes_version`.

Integration Job

`public_id`, `job_type`, `entity_type`, `entity_public_id`, `status`, `idempotency_key`, `payload_version`, `payload_hash`, `attempt_count`, `max_attempts`, `next_retry_at`, `make_execution_id`, `external_request_id`, `started_at`, `completed_at`, `last_error_code`, `last_error_message_redacted`, `result_record_type`, `result_record_public_id`.

Uniqueness is enforced logically by `idempotency_key`. Make reads the Job first; if it is `succeeded`, the repeat ends without side effects.

Audit Event

`occurred_at`, `actor_user`, `actor_role`, `action_key`, `entity_type`, `entity_public_id`, `before_json_redacted`, `after_json_redacted`, `reason`, `source` (`bubble_ui`, `bubble_backend`, `make`, `openai`), `integration_job`, `correlation_id`, `ip_hash`.

Audit Event is append-only; user workflows may not change or delete records.

6. Status transitions

Project lifecycle

```
draft → submitted → under_review
under_review → changes_requested → submitted
under_review → approved → published
under_review → rejected
published → paused | sold_out | archived
```

The admin performs `approved → published` as a separate action. Publishing is not allowed without approved verification, at least one available unit, a cover photo, a current financial snapshot, a current score, and an approved AI analysis or an explicit deterministic-only fallback.

A developer can **submit** a project with an incomplete set of units (modal: "12 units declared · 3 entered. You can submit and add the rest before publication"). So unit completeness is a condition of `approved → published`, not of `draft → submitted`. In addition, publishing requires every Financial Input used in the score to have `review_status = approved`.

Enquiry decision

Admin decisions on an introduction in the queue map to canonical statuses as follows:

UI action	Canonical transition	Emails
Approve & connect	qualified → approved_for_intro → introduced	to both parties
Hold	qualified → on_hold	none
Decline	qualified → declined	investor only, neutral text without a reason

`on_hold` and `declined` require `reason_code/note_private`. The text the investor sees on a decline comes from a template and does **not** contain the reason — the reason stays in `note_private` and the Audit Event.

`on_hold` additionally requires `follow_up_date` and `assigned_admin_user` (see FR-11 of the project specification).

Developer Application lifecycle

```

draft → submitted → under_review
under_review → more_info_required → submitted
under_review → approved | rejected
submitted/under_review → withdrawn

```

Enquiry lifecycle

```

submitted → screening → qualified → approved_for_intro → introduced
screening | qualified → on_hold | declined
on_hold → qualified | declined
introduced → developer_responded → closed_won | closed_lost
submitted | screening | qualified → cancelled

```

`qualified` means "the lead has been checked by the analyst and is waiting for the admin's decision", so it comes **before** `approved_for_intro`. The stage after the developer responds is described by `developer_responded` and then `closed_won` / `closed_lost`.

7. Privacy rules matrix

Data type	Anonymous	Investor owner	Developer company member	Admin
User	none	own whitelisted fields	own only	permitted admins
Investor Profile	none	own	none	assigned/permitted admins
Developer Company	public brand subset if approved	public subset	own company full allowed subset	full
Verification Document	none	none	own company/app	verification admins

Data type	Anonymous	Investor owner	Developer company member	Admin
Project	published public subset	published public subset	own company	full
Unit	published + available public subset	same + own saved links	own company units	full
Media Asset public	published records	published records	own company	full
Media Asset private	none	none unless specifically released	own company where allowed	full
Financial Snapshot	current public metrics only	current public metrics	own project	full
Financial Input	none	none	own project, approved only	full
Analysis Fact	approved/published subset	approved/published subset	own project published view	full
AI Analysis	approved/published subset only	approved/published subset	own project published view	full
Enquiry	none	own, investor-safe fields	own company, masked until release	full
Contact Release	none	own	own company	full
Change Request	none	none	own company	full
Audit Event / Integration Job	none	none	none	admin only

Implementation details:

- privacy conditions must check direct fields on the record itself; do not build multi-level chains for search access;
- duplicate `owner_user`, `company`, `publication_status` on protected types where that simplifies the rule;
- the workflow `Only when` checks role, membership, account status and the allowed transition; privacy rules do not replace workflow authorisation;
- do not allow auto-binding for critical fields;
- do not expose the Data API for business types publicly; Make calls narrow authenticated Workflow API endpoints.

8. Bubble workflows that maintain integrity

Workflow	Trigger	Result
<code>create_developer_application</code>	submit form	User/Company/Application without duplicates, consent, audit
<code>submit_project</code>	developer action	validation, content version, status event, review job
<code>publish_project</code>	admin action	server validation, publish unit snapshots, audit

Workflow	Trigger	Result
store_rental_comparables	callback from MK-10	Rental Comparable Set pending , previous one → is_current = no
approve_rental_comparables	admin action	sample approved , becomes available as a Financial Input source
approve_financial_input	admin action on A04	Financial Input approved , previous one → superseded_by
recalculate_unit_financials	approved price/rent/config change	new Financial Snapshot, Unit cached metrics
calculate_unit_score	current financial/model change	Listing Score + components
queue_ai_analysis	current score created	Integration Job ai_analysis_generate
rescore_all_published	admin saved a new Score Model Version	batch recalculation with progress, a new Listing Score per unit, old ones → is_current = no
submit_enquiry	investor action	consent check, Enquiry + status event
approve_introduction	admin action	Contact Release + delivery Integration Job
submit_change_request	developer action	immutable Change Items
apply_change_request	admin approval	apply whitelisted fields once, recalc, audit
close_account	confirmed request	suspend access, queue export/anonymisation review

Use database triggers only as a safety net for changes that can come from the Bubble editor/API. Main second-order updates are better called from the primary backend workflow, so workload is not spent checking every change.

9. Search and performance

- The Screen 02 search runs over Unit, with constraints: publication_status , availability , country_cached , price_eur , bedrooms_cached , gross_yield , net_yield , strategy_keys , completion_date_cached .
- Do not use :filtered for the server-filtered catalogue; all main conditions must be search constraints.
- Debounce slider changes by 300-500 ms.
- Sorting by net_yield , gross_yield , price_eur , investment_score . The completion_date_cached field is used for the completion-range filter.
- Pagination: **8 records per page**, numbered pager.
- The "Top 5" rail takes the first five records of the same sorted set — no separate query is needed.
- Do not store record lists on User if they grow without bound; Saved Unit, Enquiry, Notification are separate types.
- For dashboard counts, create an Admin Metrics Snapshot if needed rather than running dozens of live count searches on every open.

10. Retention and deletion

Retention periods are set in the platform's legal policy. MVP technical policy:

- account closure blocks login immediately; the actual delete/anonymise is a controlled backend process;
- open enquiries, consent, audit and legal/transactional records are not deleted automatically before the legal hold ends;
- investor PII in closed records is replaced with a pseudonymous reference after the approved period;
- an uploaded file is deleted by a separate action before the URL is cleared;
- the Integration Job payload must not contain full PII; store a hash and an entity reference;
- the OpenAI input snapshot contains property facts but no investor contact data.

11. Seed data before build

Required seed sets:

1. Country Config: Spain, Cyprus; Greece/Portugal as `coming_soon` if needed.
2. Cost Rules with source, effective date and approver. 2a. Data Provider: approved portals / market-data providers for Spain and Cyprus, with access method and a link to the terms of use. 2b. Market Benchmark: median price per m² for the launch districts.
3. Document Requirements per country.
4. Amenities and project/unit features.
5. Score Criteria (**five**, listed in §5.4) and the first approved Score Model Version.
6. Legal Document Versions: Terms, Privacy, Cookie, Investment Disclaimer, AI Disclaimer, Developer Agreement.
7. Notification templates and role-specific labels for Enquiry Status.

12. Migration between Bubble Development and Live

- deploy structure/option sets through Bubble version control;
- export/import reference data in a defined order using `public_id/key`;
- do not copy Live PII into Development without a written basis and masking;
- OpenAI/Make credentials are separate for Development and Live;
- smoke dataset: 2 countries, 2 companies, 4 projects, 8 unit types, 20 units, all availability/status cases.

13. Architecture acceptance criteria

- no anonymous request receives an email, phone, exact private address or private document URL;
- before introduction the developer sees the lead without PII; after approval sees only whitelisted contact fields;
- a repeated Make webhook with the same idempotency key creates no duplicate and sends no repeat email;
- every published unit has a current Financial Snapshot, Listing Score and Score Model Version;

- every number on Analysis is traceable to a specific snapshot/source key;
- score components equal the total score within 0.01, and there are exactly five;
- a change to an approved price creates a Change Request, an Audit Event, a new Financial Snapshot and a new Listing Score;
- an availability change likewise goes through a Change Request and is not applied before approval;
- an anonymous request does not receive `Project.company` or any developer field;
- no Financial Input with `source = developer_claim` takes part in scoring without separate approval;
- every published unit has at least one Analysis Fact or an explicit note that there are no gaps;
- deleting a private record does not leave an accessible attached file;
- every status change creates a Status Event/Audit Event;
- admin publish and contact release are impossible without server-side authorisation.